UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 005 · Particular integral Type 3 (X = x^m) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·8	Unit 2 deck 005 (“Particular integral of Type 3”), slide 8 (rendered .pptx)
T005 00:06:48	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 3” (12.6 min), segment start
T006a 00:06:05	Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min), its Type 3 part 00:00:00–00:09:55; the class 006 pack cites the whole file as T006b
CW U2 P15	Unit 2 class-work problem 15
ESA Jul-23 3b	Past paper: UE22MA141A Jul 2023, question 3.b
ESA Jan-24 s2 2d	Past paper: UE23MA141A Jan 2024 set 2, question 2.d

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A Formula sheet

1 The Type 3 rule: $X = x^m$

E1.1 · Type 3 PI

ESA ★★★

$$y_p = \text{P.I} = \frac{1}{f(D)}x^m = \big[f(D)\big]^{-1}x^m$$

GIVES The PI of a power of x (or a polynomial).

APPLIES $X = x^m$, m a positive integer.

NOTE The PI is a polynomial of degree m when $f(0) \neq 0$.

S005·4 · S005·10 · T005 00:00:50 · T006a 00:00:36

E1.2 · Take the lowest-degree term out

ESA ★★★

$$\frac{1}{f(D)}x^m = \frac{1}{a_0\,[1 \pm \varphi(D)]}x^m = \frac{1}{a_0}\big[1 \pm \varphi(D)\big]^{-1}x^m$$

GIVES A bracket ready for the binomial series.

APPLIES a_0 = the lowest-degree term (usually the constant).

NOTE Take out the constant with its sign: $D^2 - 1 = -(1 - D^2)$, $D^2 + 5D + 4 = 4\big[1 + \frac{D^2+5D}{4}\big]$.

S005·4 · S005·7 · S005·8 · T005 00:01:08 · T005 00:09:16

E1.3 · Powers of D on x^m

ESA ★★★

$$D^k x^m = \frac{m!}{(m-k)!}\,x^{m-k}\ (k \le m), \qquad D^k x^m = 0\ (k > m)$$

GIVES Why the series stops at D^m; $D^m x^m = m!$ is a constant.

APPLIES Operating term by term, $D = \frac{d}{dx}$.

NOTE Example: $D(x^2 + 7x + 9) = 2x + 7$, $D^2(x^2 + 7x + 9) = 2$, $D^3(\dots) = 0$.

S005·4 · S005·10 · T005 00:01:51 · T006a 00:01:29 · T006a 00:02:06

2 The binomial expansions and where to stop

E2.1 · First-power series

ESA ★★★

$$(1+z)^{-1} = 1 - z + z^2 - z^3 + \dots$$
$$(1-z)^{-1} = 1 + z + z^2 + z^3 + \dots$$

GIVES $[1 \pm \varphi(D)]^{-1}$ with $z = \varphi(D)$.

APPLIES After E1.2.

NOTE Signs alternate for $(1+z)^{-1}$; all plus for $(1-z)^{-1}$ (replace z by $-z$).

S005·5 · T005 00:02:45 · T005 00:03:12 · T006a 00:04:36

E2.2 · Second-power series

ESA ★★

$$(1+z)^{-2} = 1 - 2z + 3z^2 - 4z^3 + \dots$$
$$(1-z)^{-2} = 1 + 2z + 3z^2 + 4z^3 + \dots$$

GIVES $[1 \pm \varphi(D)]^{-2}$ when $f(D)$ is a square, e.g. $(D-2)^2$, $(D^2-1)^2$.

APPLIES $f(D) = a_0^2[1 \pm \varphi(D)]^2$.

NOTE "If we remember the first and the third, it is sufficient."

S005·5 · T005 00:03:46 · T005 00:04:04 · CW U2 P21 · CW U2 P22

E2.3 · Keep only powers up to D^m

ESA ★★★

$$(D^2+3D)^2 = D^4 + 6D^3 + 9D^2 \;\to\; 6D^3 + 9D^2$$
$$(D^2+3D)^3 = D^6 + \dots + 27D^3 \;\to\; 27D^3 \qquad (m=3)$$

GIVES The terms of the square and cube that survive.

APPLIES $\varphi(D)$ has two or more terms.

NOTE Lecture: "we don't want a^2, we want b^2, ... $2ab$ that is $6D^3$".

S005·12 · T006a 00:06:05 · T006a 00:06:22 · T006a 00:06:43 · T005 00:11:24

3 When D is a factor of $f(D)$

E3.1 · D as a factor

ESA ★

$$\frac{1}{D^k\,\psi(D)}x^m = \frac{1}{D^k}\Big[\big(\psi(D)\big)^{-1}x^m\Big], \qquad \frac{1}{D} = \int (\,)\,dx$$

GIVES The PI when $f(D)$ has no constant term.

APPLIES $f(D) = D^k\psi(D)$, $\psi(0) \neq 0$.

NOTE Expand $\psi(D)^{-1}$ up to D^m, operate, then integrate k times (no constants). The PI has degree $m + k$.

ADDED FOR CLARITY S005·4

E3.2 · A constant right-hand side

ESA ★★

$$\frac{1}{f(D)}k = \frac{k}{f(0)}\ (f(0) \neq 0), \qquad \frac{1}{D\,\psi(D)}k = \frac{kx}{\psi(0)}$$

GIVES The PI of the constant term of a polynomial or of $\sin^2$, $\cos^2$.

APPLIES $m = 0$: $k = kx^0 = ke^{0x}$.

NOTE Type 1 with $a = 0$ (class 003) and Type 3 with $m = 0$ agree.

ADDED FOR CLARITY ESA Jan-24 s4 2c

4 Mixed right-hand sides: Types 1, 2 and 3 together

E4.1 · Superposition

ESA ★★★

$$\frac{1}{f(D)}(X_1 + X_2 + X_3) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 + \frac{1}{f(D)}X_3 = \mathrm{PI}_1 + \mathrm{PI}_2 + \mathrm{PI}_3$$

GIVES The PI of a mixed right-hand side.

APPLIES X is a sum of terms of different types.

NOTE Each term chooses its own rule and its own failure test.

ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c · CW U2 P21

E4.2 · Reminder: e^{ax} and $a^{\pm x}$ (Type 1)

ESA ★★★

$$\frac{e^{ax}}{f(D)} = \frac{e^{ax}}{f(a)}, \qquad \frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)} \qquad (f \neq 0)$$

GIVES PI of the exponential terms; on failure $x\frac{e^{ax}}{f'(a)}$, then $x^2\frac{e^{ax}}{f''(a)}$.

APPLIES $a^{-x} = e^{-x\log a}$, so $D \to -\log a$.

NOTE Taught in the class 003 pack; 2^{-x} is in two ESA questions.

ESA Jan-24 s1 2d · ESA Jan-24 s4 2c · ESA Jan-24 s2 2d

E4.3 · Reminder: $\sin ax$, $\cos ax$ (Type 2)

ESA ★★★

$$\frac{\sin ax}{f(D^2)} = \frac{\sin ax}{f(-a^2)}, \qquad \frac{\cos ax}{D^2 + a^2} = \frac{x \sin ax}{2a}$$

GIVES PI of the trigonometric terms, with the failure case.

APPLIES Replace D^2 by $-a^2$; rationalise a lone D.

NOTE Taught in the class 004 pack.

ESA Jul-23 3b · ESA Jan-24 s1 2d · CW U2 P21

B Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.3, E3.1
$f(D)$	Operator polynomial	Left-hand side operator of $f(D)y = X$	E1.1
a_0	Lowest-degree term	Taken out of $f(D)$ with its sign (usually the constant term)	E1.2
$\varphi(D)$	Phi of D	Rest of $f(D)$ after a_0 is taken out; only positive powers of D	E1.2
$\psi(D)$	Psi of D	$f(D)/D^k$ when D^k is a factor; $\psi(0) \neq 0$	E3.1
z	Series variable	Stands for $\varphi(D)$; slide 005·5 writes it as x (not the variable x)	E2.1, E2.2
m	Degree of X	Expand up to D^m; slide 005·4 calls it n. Also the AE unknown (same letter)	E1.1, E1.3
k	Power of the factor D	Number of integrations; also a constant in E3.2	E3.1, E3.2
PI_1, PI_2, PI_3	Partial PIs	PIs of the separate terms of X	E4.1
X	Right-hand side	A function of x; capital X, small x	E1.1

Remark. The slides use n (slide 005·4) and m (slide 005·10) for the same thing, the degree of the polynomial $X = x^n$ or x^m; this pack uses m. The letter m is also the unknown of the auxiliary equation ($m^2 - 1 = 0$): the two never appear in the same line. $\varphi(D)$ is "phi of D", the rest of $f(D)$ after the lowest-degree term is taken out; in the series of slide 005·5 the letter x stands for $\varphi(D)$, not for the variable x (this pack writes z there).